

Introduction to the Free Energy Principle

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Preface

This is a Quarto book.

To learn more about Quarto books visit <https://quarto.org/docs/books>.

1 Introduction

This is a book created from markdown and executable code.

See Knuth (1984) for additional discussion of literate programming.

2 From path surprisal to least action

Understanding Equation (4) of The free energy principle made simpler but not too simple

2.1 The question we want to answer

A bead in water is pulled toward a resting position while surrounding molecules give it random kicks. Release it repeatedly from the same point: each run draws a different curve. Can we identify a path with the highest density, and why should that path obey the noiseless equation of motion?

This chapter builds Equation (4) of Friston et al. (2023) from probability, then explains each of its three lines:

$$\begin{aligned}\mathbf{x}[\tau] &= \arg \min_{x[\tau]} \mathcal{A}(x[\tau]) \\ \Leftrightarrow \delta_x \mathcal{A}(\mathbf{x}[\tau]) &= 0 \\ \Leftrightarrow \dot{\mathbf{x}}(\tau) &= f(\mathbf{x}).\end{aligned}\tag{4}$$

The bold type denotes the selected most likely path, not simply a vector. Here x may itself be a vector.

Prerequisites: derivatives, Gaussian densities, integration, and matrix multiplication. We introduce functionals and variations as we need them. The numerical sections use NumPy, SciPy, and Matplotlib; all plots are saved inside this notebook after execution. You can read the derivation without running Python.

Our working assumptions: additive noise with constant symmetric positive-definite Γ ; fixed initial state $x(0) = x_0$; free terminal state; sufficiently regular drift with a unique deterministic solution over $[0, \tau]$. For the central proof, we omit the divergence term exactly as the paper's footnote 4 does. We will explain that qualification rather than silently extending the result to every stochastic system.

The learning route is: **step probabilities** $\rightarrow$ **path surprisal** $\rightarrow$ **quadratic action** $\rightarrow$ **its minimum** $\rightarrow$ **vanishing variation** $\rightarrow$ **deterministic flow**.

2.2 1. A snapshot is different from a movie

The paper starts with the formal Langevin equation

$$\dot{x}(t) = f(x(t)) + \omega(t).$$

Here f is the systematic drift and ω represents rapidly fluctuating noise. For our spring-like example, $f(x) = -kx$, with $k > 0$.

Symbol	Meaning	Bead example
$x(t)$	State at one time	One photograph of its position
$x[\tau] = \{x(t) : 0 \leq t \leq \tau\}$	Whole trajectory	The complete movie
$p(x, \tau)$	State density	Where it may be at time τ
$p(x[\tau] x_0)$	Formal path density given its start	Which movie may unfold

The Fokker–Planck equation, Equation (2), evolves the state density. Equation (3) describes paths through their action. Equation (4) selects a path from that description. Equation (5) names the negative log state density *surprisal* and relates average surprisal and action to entropies.

In continuous spaces, an exact state or exact path normally has probability zero. A density describes relative likelihood with respect to a specified measure. We first use a finite time grid, where the density over $(x_1, \dots, x_N)$ is unambiguous.

2.3 2. Surprisal is the right currency

For a discrete event E , define

$$\mathfrak{I}(E) = -\ln p(E).$$

An event of probability 1 has zero surprisal; an event of probability 0.01 has surprisal about 4.605 nats. The natural logarithm gives the unit *nat*. Rare events receive larger scores.

The crucial property is

$$-\ln(ab) = -\ln a - \ln b.$$

For independent events, probabilities multiply and surprisals add. For dependent events we instead use conditional probabilities:

$$-\ln p(A, B) = -\ln p(A) - \ln p(B | A).$$

This conditional version is what a trajectory needs. **Successive positions are not independent.** Their dependence is retained by conditioning each step on the previous position.

The paper defines action by applying the same transformation to a whole path:

$$\mathcal{A}(x[\tau]) \equiv -\ln p(x[\tau] | x_0).$$

Thus “action is path surprisal” is a definition. The work is to show why it has the integral structure of an action.

A negative log density need not be positive and changes with coordinate units. Our later positivity argument concerns the **quadratic part after removing a path-independent constant**, not every negative log density.

2.4 3. Make white noise precise with time increments

A useful rigorous version of Equation (1) is

$$dx_t = f(x_t) dt + \sqrt{2\Gamma} dW_t,$$

where W_t is an n -dimensional Wiener process and the matrix square root obeys $\sqrt{2\Gamma}\sqrt{2\Gamma}^T = 2\Gamma$.

Divide $[0, \tau]$ into N steps of width $\Delta t = \tau/N$. The Euler–Maruyama approximation is

$$x_{j+1} = x_j + f(x_j)\Delta t + \sqrt{2\Gamma\Delta t}\xi_j, \quad \xi_j \sim \mathcal{N}(0, I),$$

with independent ξ_j . Therefore,

$$p_{\Delta t}(x_{j+1} | x_j) = \mathcal{N}(x_{j+1}; x_j + f(x_j)\Delta t, 2\Gamma\Delta t).$$

This is exact for the discretized model and a small-step approximation to the diffusion’s transitions.

Why the Δt matters: the random *increment* has covariance $2\Gamma\Delta t$. The approximate velocity $(x_{j+1} - x_j)/\Delta t$ has covariance $2\Gamma/\Delta t$, which diverges as the time step shrinks. White noise has

no ordinary finite-variance value at an instant, and typical diffusion paths are not differentiable. We should not literally integrate $-\ln p(\dot{x} \mid x)$ as though it were an ordinary finite surprisal rate.

We use smooth candidate paths to derive the action, and discrete random paths for the simulations.

2.5 4. Multiply the step densities, then take a logarithm

The probability chain rule, followed by the Markov property, gives

$$p_{\Delta t}(x_1, \dots, x_N \mid x_0) = \prod_{j=0}^{N-1} p_{\Delta t}(x_{j+1} \mid x_j).$$

Define the unexplained increment

$$r_j = x_{j+1} - x_j - f(x_j)\Delta t.$$

It is the random kick that this candidate path requires. The Gaussian formula gives

$$-\ln p_{\Delta t}(x_{j+1} \mid x_j) = \frac{1}{2} \ln[(4\pi\Delta t)^n |\Gamma|] + \frac{1}{4\Delta t} r_j^\top \Gamma^{-1} r_j.$$

Summing over steps:

$$\mathcal{A}_{\Delta t} = C_{\Delta t} + S_{\Delta t},$$

$$C_{\Delta t} = \frac{N}{2} \ln[(4\pi\Delta t)^n |\Gamma|], \quad S_{\Delta t} = \sum_{j=0}^{N-1} \frac{r_j^\top \Gamma^{-1} r_j}{4\Delta t}.$$

Every candidate path on the same grid receives the same $C_{\Delta t}$. It therefore drops out of minimization and density ratios:

$$\frac{p_{\Delta t}(x \mid x_0)}{p_{\Delta t}(y \mid x_0)} = \exp[-(S_{\Delta t}[x] - S_{\Delta t}[y])].$$

Do not compare absolute discrete densities across different grids: their dimensions and normalizers change.

2.6 5. The sum becomes a quadratic action

For a smooth candidate path,

$$r_j \approx [\dot{x}(t_j) - f(x(t_j))]\Delta t.$$

Substituting into $S_{\Delta t}$ leaves one factor of Δt , so the sum approaches

$$S[x] = \int_0^\tau \mathcal{L}_0(x, \dot{x}) dt, \quad \mathcal{L}_0 = \frac{1}{4}(\dot{x} - f)^\top \Gamma^{-1}(\dot{x} - f).$$

Define the drift residual

$$\varepsilon(t) = \dot{x}(t) - f(x(t)).$$

Then $\mathcal{L}_0 = \frac{1}{4}\varepsilon^\top \Gamma^{-1}\varepsilon$ is the instantaneous quadratic cost of the forcing needed to produce the path. It is sometimes tempting to call ε a “prediction error”; here it is simply a physical mismatch with the drift, without assuming a cognitive prediction.

Intuition: a path that follows the flow requires no extra forcing. A path that fights the flow needs sustained kicks, and its cost accumulates. The matrix Γ^{-1} weights directions: a deviation along a direction with weak noise costs more than one along a direction with strong noise.

The paper’s Equation (3) writes a formal continuum normalizer and, with its overall factor of one half, the Lagrangian

$$\mathcal{L} = \frac{1}{4}\varepsilon^\top \Gamma^{-1}\varepsilon + \frac{1}{2}\nabla \cdot f.$$

The $\frac{1}{2}\nabla \cdot f$ term comes from a Jacobian in the Stratonovich path construction, not from the Gaussian exponent alone. Different path-measure conventions require care. The discrete normalizer above should **not** be claimed to converge to a finite continuum constant. For Equation (4), what we need is the path-dependent quadratic part with the divergence contribution omitted.

2.7 6. First line: what kind of object are we minimizing?

An ordinary function takes a number or vector and returns a number. A **functional** takes a whole function and returns a number:

$$x[\tau] \mapsto S[x].$$

Imagine many curves drawn on transparent sheets. For each sheet, evaluate one score: its total quadratic cost. The **argmin** returns the winning **curve**. The **min** would return its **score**.

Because $-\ln$ is decreasing,

$$\arg \min_{x[\tau]} \mathcal{A}[x] = \arg \max_{x[\tau]} p(x[\tau] \mid x_0),$$

under the chosen density convention. This likelihood interpretation is built into the definition; it does not require a new physical principle. It should not be confused with identifying this stochastic action with the mechanical action $\int (T - V) dt$.

Only the starting point is fixed. The terminal point is allowed to be whichever point the minimizing path reaches. That freedom will be decisive.

2.8 7. First line to third line: a proof without variational calculus

Because Γ is positive definite,

$$\mathcal{L}_0 \geq 0, \quad \mathcal{L}_0 = 0 \iff \varepsilon = 0.$$

Consequently $S[x] \geq 0$ for every admissible path. Now solve the noiseless initial-value problem

$$\dot{\mathbf{x}}(t) = f(\mathbf{x}(t)), \quad \mathbf{x}(0) = x_0.$$

Under our assumptions this solution exists on the interval and is unique. It has $\varepsilon(t) = 0$ throughout, so it attains $S[\mathbf{x}] = 0$. Nothing can beat zero.

Conversely, attaining zero requires $\varepsilon = 0$ almost everywhere (and everywhere for smooth residuals). We have proved

$\text{global minimum of } S \iff \dot{\mathbf{x}} = f(\mathbf{x}).$

A useful precision: $\mathcal{L}_0$ is quadratic in ε , but $\varepsilon = \dot{x} - f(x)$ need not be linear in x . For nonlinear drift, the action need not be convex as a functional of the path. Our proof uses a nonnegative lower bound and an admissible path that attains it; it does not assume global convexity.

2.9 8. Middle line: what is a variation?

We now check why the winning path has $\delta_x \mathcal{A} = 0$. Deform a candidate path by a small amount:

$$x_h(t) = x(t) + h\eta(t).$$

Here h is a small scalar and $\eta(t)$ is a whole deformation shape. It must satisfy $\eta(0) = 0$ because the initial state is fixed, but $\eta(\tau)$ is unrestricted.

The **first variation** is the first-order change in the score:

$$\delta S[x; \eta] = \left. \frac{d}{dh} S[x + h\eta] \right|_{h=0}.$$

Saying $\delta_x S = 0$ means that this derivative vanishes for every admissible deformation. This generalizes setting $dg/dx = 0$ in ordinary calculus.

For constant Γ , let $J_{ij} = \partial f_i / \partial x_j$. Then

$$\delta \varepsilon = \dot{\eta} - J\eta,$$

and therefore

$$\delta S = \frac{1}{2} \int_0^\tau \varepsilon^\top \Gamma^{-1} (\dot{\eta} - J\eta) dt.$$

Integrating the $\dot{\eta}$ term by parts gives

$$\delta S = \left[\frac{1}{2} \varepsilon^\top \Gamma^{-1} \eta \right]_0^\tau - \frac{1}{2} \int_0^\tau \eta^\top (\Gamma^{-1} \dot{\varepsilon} + J^\top \Gamma^{-1} \varepsilon) dt.$$

This has two parts: an interior equation and a boundary term. We must satisfy both.

2.10 9. The free endpoint completes the argument

First choose deformations that vanish at both ends. Their interior shape is arbitrary, so stationarity requires

$$\Gamma^{-1}\dot{\varepsilon} + J^T\Gamma^{-1}\varepsilon = 0,$$

or

$$\boxed{\dot{\varepsilon} = -\Gamma J^T\Gamma^{-1}\varepsilon.}$$

This is the Euler–Lagrange equation in residual form. Along any given candidate path its coefficients are functions of time and the equation is linear and homogeneous in ε . **By itself it does not force $\varepsilon = 0$.**

Next allow the final deformation $\eta(\tau)$ to be arbitrary. The boundary term must vanish for every such choice:

$$\left. \frac{\partial \mathcal{L}_0}{\partial \dot{x}} \right|_{\tau} = \frac{1}{2}\Gamma^{-1}\varepsilon(\tau) = 0.$$

This is the **natural boundary condition**. It says $\varepsilon(\tau) = 0$. Uniqueness for the homogeneous residual equation, solved backward from that terminal value, implies $\varepsilon(t) = 0$ throughout.

For $f(x) = -kx$ in one dimension, this is especially transparent:

$$\dot{\varepsilon} = k\varepsilon, \quad \varepsilon(t) = ce^{kt}, \quad \varepsilon(\tau) = 0 \Rightarrow c = 0.$$

Thus, for this simplified problem and these admissible variations, stationarity selects the zero-cost global minimum. In a general variational problem, stationarity alone does not guarantee a minimum.

If both endpoints were fixed, $\eta(\tau) = 0$ automatically and the natural boundary condition would disappear. A required terminal point may be unreachable by the drift alone. The least-cost path would then need nonzero forcing.

2.11 10. Execute Equation (4): directly minimize a discrete path

Let $f(x) = -kx$. Represent a path by the vector of unknown future positions $(x_1, \dots, x_N)$ and minimize $S_{\Delta t}$ with x_0 fixed. Every residual is linear in this vector, so this example is a least-squares problem. SciPy's `least_squares` minimizes one half of the residual norm squared; our scaling makes its objective exactly $S_{\Delta t}$.

We compare the optimized path with both the Euler drift recursion and the exact ODE solution. The optimizer should agree with the **Euler path on the same grid**; the difference from the exact exponential is a time-discretization error.

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.optimize import least_squares

plt.rcParams.update({"figure.dpi": 120, "font.size": 11,
                    "axes.spines.top": False, "axes.spines.right": False})
k, gamma, x0, tau, N = 1.0, 0.20, 1.5, 3.0, 80
dt = tau / N
t = np.linspace(0.0, tau, N + 1)
a = 1.0 - k * dt

def discrete_action(path, drift, diffusion, step):
    """Return the quadratic Euler action along the last array axis.

    Args:
        path: One path or a batch of equally spaced paths.
        drift: Function acting on the current positions.
        diffusion: Positive scalar Gamma.
        step: Positive time-step duration.

    Returns:
        Quadratic action, excluding the Gaussian normalizer.
    """
    residual = np.diff(path, axis=-1) - drift(path[..., :-1]) * step
    return np.sum(residual**2, axis=-1) / (4.0 * diffusion * step)

def scaled_residuals(future_positions):
    """Return residuals scaled for SciPy's half-squared-norm objective."""
    path = np.r_[x0, future_positions]
    return (np.diff(path) + k * path[:-1] * dt) / np.sqrt(2 * gamma * dt)

initial_guess = np.linspace(x0, -0.5, N + 1)
```

```

result = least_squares(scaled_residuals, initial_guess[1:],
                      gtol=1e-11, ftol=1e-11, xtol=1e-11)
optimized = np.r_[x0, result.x]
euler_mode = x0 * a ** np.arange(N + 1)
exact_ode = x0 * np.exp(-k * t)
assert result.success, result.message
assert np.max(np.abs(optimized - euler_mode)) < 1e-7
assert discrete_action(optimized, lambda x: -k*x, gamma, dt) < 1e-12
print(f"Initial action: {discrete_action(initial_guess, lambda x: -k*x, gamma, dt):.4f}")
print(f"Optimized action: {result.cost:.3e}")
print(f"Maximum difference from Euler drift path: {np.max(np.abs(optimized-euler_mode)):.3e}")
print(f"Maximum Euler-versus-exact ODE error: {np.max(np.abs(euler_mode-exact_ode)):.4f}")

```

Initial action: 1.3389

Optimized action: 4.219e-30

Maximum difference from Euler drift path: 3.331e-16

Maximum Euler-versus-exact ODE error: 0.0105

```

fig, axes = plt.subplots(1, 2, figsize=(11, 4), constrained_layout=True)
axes[0].plot(t, initial_guess, color="0.65", linestyle="--", label="Initial guess")
axes[0].plot(t, exact_ode, color="#d97706", label="Exact noiseless ODE")
axes[0].plot(t, optimized, color="#087e8b", linestyle=":", linewidth=3,
             label="Optimized discrete path")
axes[0].set(xlabel="Time", ylabel="Position", title="A whole path is the argument")
axes[0].legend(fontsize=9)
# This deformation preserves the start and leaves the endpoint free.
amplitudes = np.linspace(-1.5, 1.5, 101)
shape = t / tau
family = euler_mode[None, :] + amplitudes[:, None] * shape[None, :]
costs = discrete_action(family, lambda x: -k*x, gamma, dt)
axes[1].plot(amplitudes, costs, color="#087e8b")
axes[1].scatter([0], [0], color="#d97706", zorder=3)
axes[1].set(xlabel="Deformation amplitude h", ylabel="Quadratic action",
            title="One slice through the space of paths")
plt.show()

```

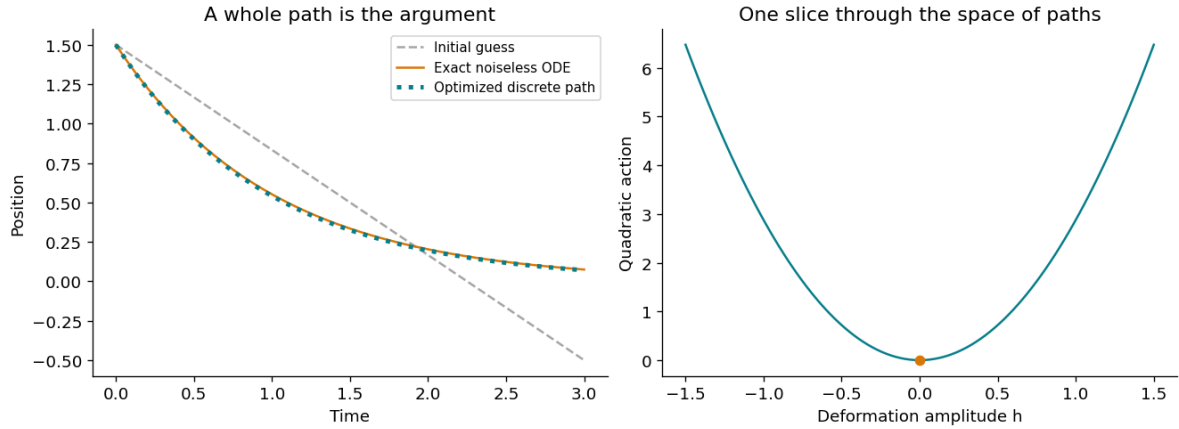


Figure 2.1: Minimizing the discrete action recovers the drift recursion. Moving away from it along a chosen deformation raises the action quadratically.

Read the output: the minimum score is essentially zero, up to numerical precision. The right panel varies just one deformation shape; the analytic nonnegativity proof establishes the minimum over all admissible paths, not only the plotted family.

2.11.1 Change the question: force a different endpoint

For the linear example, the continuum minimizer with $x(\tau) = b$ is

$$x_b(t) = x_0 e^{-kt} + \frac{b - x_0 e^{-k\tau}}{\sinh(k\tau)} \sinh(kt).$$

Its residual is proportional to e^{kt} , just as the Euler–Lagrange equation predicts. Unless $b = x_0 e^{-k\tau}$, it has positive cost. The next cell solves the analogous discrete problem while keeping both ends fixed.

```
b = -0.8

def bridge_residuals(interior):
    """Scale residuals for a path whose two endpoints are fixed."""
    path = np.r_[x0, interior, b]
    return (np.diff(path) + k * path[:-1] * dt) / np.sqrt(2*gamma*dt)

bridge_result = least_squares(bridge_residuals, np.linspace(x0, b, N+1)[1:-1],
                              gtol=1e-11, ftol=1e-11, xtol=1e-11)
bridge = np.r_[x0, bridge_result.x, b]
```



```

bridge_continuum = exact_ode + (b-exact_ode[-1])*np.sinh(k*t)/np.sinh(k*tau)
# Independent analytic solution for the discrete linear bridge.
D = b - euler_mode[-1]
weights = a ** np.arange(N-1, -1, -1)
r_star = D * weights / np.dot(weights, weights)
bridge_exact_discrete = np.empty(N+1)
bridge_exact_discrete[0] = x0
for j in range(N):
    bridge_exact_discrete[j+1] = a*bridge_exact_discrete[j] + r_star[j]
assert bridge_result.success
assert np.max(np.abs(bridge - bridge_exact_discrete)) < 1e-7
assert bridge_result.cost > 0
fig, ax = plt.subplots(figsize=(7, 4), constrained_layout=True)
ax.plot(t, euler_mode, label="Free endpoint", color="#087e8b")
ax.plot(t, bridge, label="Fixed endpoint: discrete optimum", color="#a23b72")
ax.plot(t, bridge_continuum, "--", label="Fixed endpoint: continuum solution", color="#d97700")
ax.scatter([tau], [b], color="#a23b72", zorder=3)
ax.set(xlabel="Time", ylabel="Position", title="Boundary conditions decide which paths compete")
ax.legend(fontsize=9)
plt.show()
print(f"Action with imposed endpoint: {bridge_result.cost:.4f}")

```

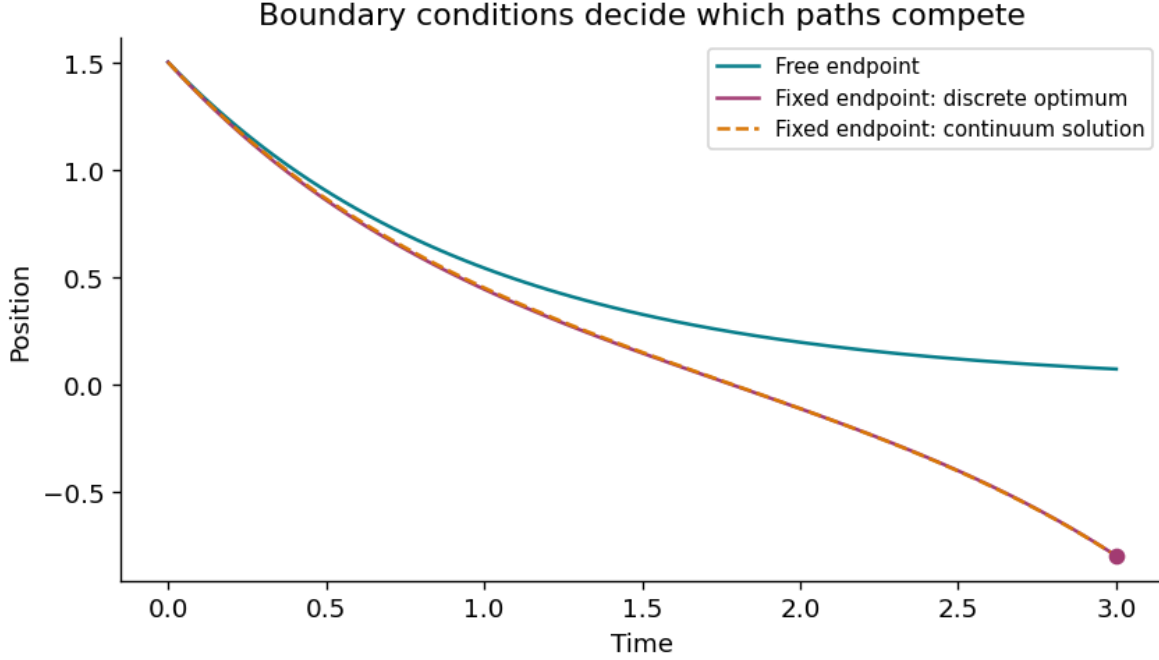


Figure 2.2: A forced endpoint changes the variational problem: the drift path is no longer admissible.

Action with imposed endpoint: 1.8630

2.12 11. A most likely path is not a typical noisy realization

At a fixed grid resolution, sample the Euler stochastic dynamics. For each path, the increments satisfy $r_j = \sqrt{2\gamma\Delta t}\xi_j$. Therefore

$$S_{\Delta t} = \frac{1}{2} \sum_{j=0}^{N-1} \xi_j^2, \quad 2S_{\Delta t} \sim \chi_N^2.$$

The expected action is $N/2$, whereas the modal path has action zero. There is no contradiction: a high-dimensional Gaussian has its density maximum at its center, but many more configurations lie away from that center.

An exactly noiseless path has probability zero for nonzero diffusion, as does every exact path. The finite-grid modal path remains meaningful as a density maximum. In continuous time, “most likely path” requires a path measure or small-tube convention; it should not be mistaken for an event of positive probability.

```

rng = np.random.default_rng(20260906)
M = 3000
samples = np.empty((M, N+1))
samples[:, 0] = x0
for j in range(N):
    samples[:, j+1] = a*samples[:, j] + np.sqrt(2*gamma*dt)*rng.standard_normal(M)
actions = discrete_action(samples, lambda x: -k*x, gamma, dt)
fig, axes = plt.subplots(1, 2, figsize=(11, 4), constrained_layout=True)
axes[0].plot(t, samples[:,25].T, color="#087e8b", alpha=0.18, linewidth=0.8)
axes[0].plot(t, euler_mode, color="black", linewidth=2, label="Modal discrete path")
axes[0].set(xlabel="Time", ylabel="Position", title="Mode and noisy samples")
axes[0].legend(fontsize=9)
axes[1].hist(actions, bins=40, color="#087e8b", alpha=0.8)
axes[1].axvline(0, color="black", label="Modal path: S = 0")
axes[1].axvline(N/2, color="#d97706", linestyle="--", label="Expected sample action: N/2")
axes[1].set(xlabel="Discrete quadratic action", ylabel="Number of paths", title="Density peak")
axes[1].legend(fontsize=9)
plt.show()
# The sample mean has standard error sqrt(N/(2M)).
se_action = np.sqrt(N/(2*M))
assert abs(actions.mean()-N/2) < 6*se_action
print(f"Mean sample action: {actions.mean():.2f}; theory: {N/2:.2f}")

```

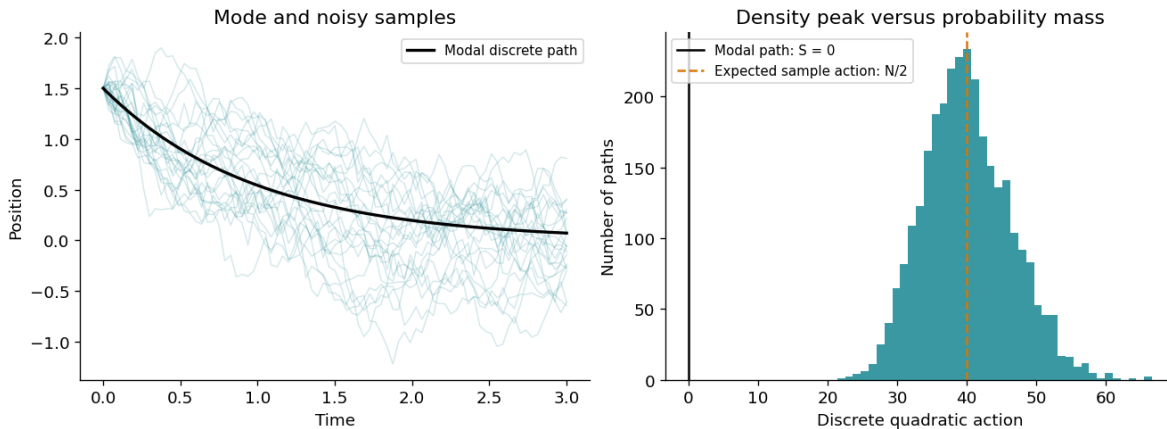


Figure 2.3: Individual noisy paths fluctuate around the drift trajectory, while their discrete actions concentrate far above the modal path's zero action.

Mean sample action: 40.06; theory: 40.00

Why specify a grid? Increasing N raises the expected sample action $N/2$. Typical Brownian paths do not have finite classical squared-velocity action in the continuum limit. Smooth-path action minimization and sampling rough diffusion trajectories are related constructions, but they are not interchangeable pointwise calculations.

2.13 12. The drift path is not generally the ensemble mean

For additive noise with zero mean,

$$\frac{d}{dt}\mathbb{E}[x_t] = \mathbb{E}[f(x_t)].$$

If $f(x) = -kx$, linearity gives $\mathbb{E}[f(x)] = f(\mathbb{E}[x])$. The mean follows the noiseless ODE. For nonlinear f , that equality generally fails.

Use a double-well drift $f(x) = x - x^3$. A deterministic trajectory starting at $x_0 = 0.5$ approaches $+1$. Noise can carry some realizations into the left well, and the ensemble mean can move away from the deterministic curve. Here we compare the mean with the **noiseless reference**, not assert that the latter is the exact finite-noise Onsager–Machlup mode: this nonlinear flow has a nonconstant divergence.

```
rng_nl = np.random.default_rng(314159)
T_nl, dt_nl, gamma_nl, M_nl = 5.0, 0.002, 0.30, 6000
steps_nl = round(T_nl/dt_nl)
time_nl = np.linspace(0, T_nl, steps_nl+1)
state = np.full(M_nl, 0.5)
mean_nl = np.empty(steps_nl+1)
se_nl = np.empty_like(mean_nl)
drift_path = np.empty_like(mean_nl)
mean_nl[0], se_nl[0], drift_path[0] = 0.5, 0.0, 0.5
for j in range(steps_nl):
    state += (state-state**3)*dt_nl + np.sqrt(2*gamma_nl*dt_nl)*rng_nl.standard_normal(M_nl)
    mean_nl[j+1] = state.mean()
    se_nl[j+1] = state.std(ddof=1)/np.sqrt(M_nl)
    y = drift_path[j]
    drift_path[j+1] = y + (y-y**3)*dt_nl
assert np.isfinite(state).all()
assert abs(mean_nl[-1]-drift_path[-1]) > 6*se_nl[-1]
fig, ax = plt.subplots(figsize=(7, 4), constrained_layout=True)
ax.plot(time_nl, drift_path, color="black", label="Noiseless reference")
ax.plot(time_nl, mean_nl, color="#a23b72", label="Ensemble mean")
ax.fill_between(time_nl, mean_nl-1.96*se_nl, mean_nl+1.96*se_nl,
```

```

        color="#a23b72", alpha=0.2, label="Approx. 95% Monte Carlo interval")
ax.set(xlabel="Time", ylabel="Position", title="Averaging a nonlinear drift changes the answer")
ax.legend(fontsize=9)
plt.show()
print(f"Terminal mean: {mean_nl[-1]:.3f} ± {1.96*se_nl[-1]:.3f} (Monte Carlo only)")
print(f"Terminal noiseless position: {drift_path[-1]:.3f}")

```

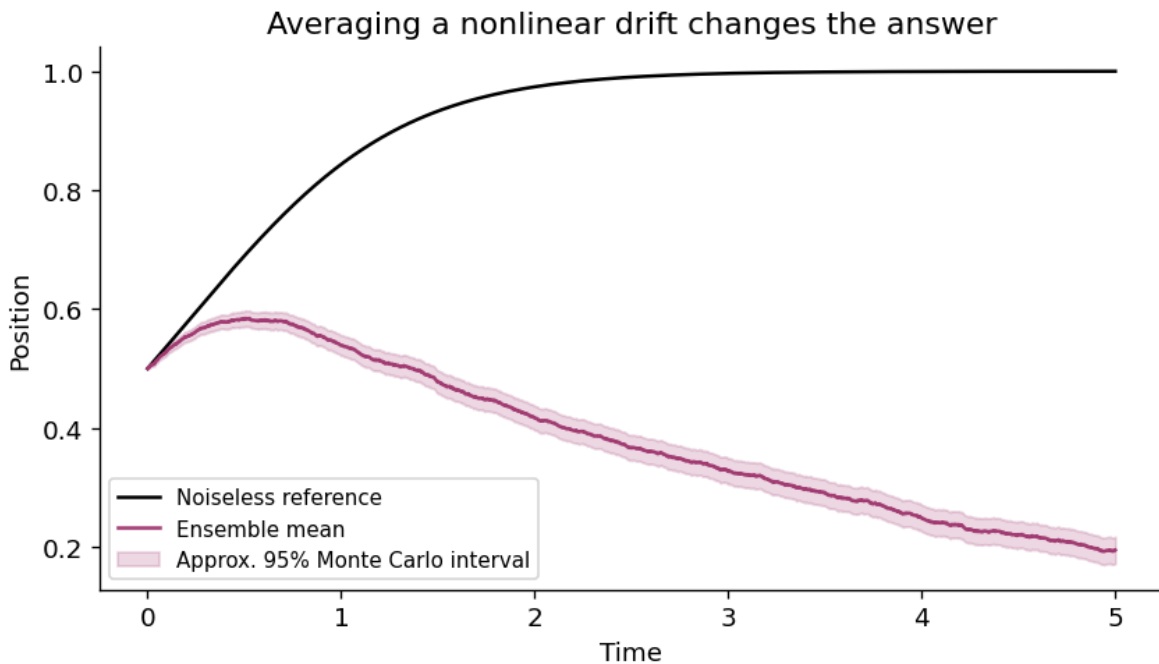


Figure 2.4: For a nonlinear double-well drift, the ensemble mean differs from the noiseless trajectory. The shaded band shows Monte Carlo uncertainty in the mean, not the spread of individual paths.

```

Terminal mean: 0.195 ± 0.023 (Monte Carlo only)
Terminal noiseless position: 1.000

```

2.14 13. Return to the divergence term

The divergence

$$\nabla \cdot f = \sum_i \frac{\partial f_i}{\partial x_i}$$

measures local expansion or contraction of a small cloud of deterministic states. The paper includes $\frac{1}{2}\nabla \cdot f$ in its full Lagrangian and explicitly omits its contribution in the explanation of Equation (4).

For constant divergence, the extra integral is the same for every path of a fixed duration. Our linear example has divergence $-k$, so the minimizer is unaffected.

For nonconstant divergence, its variation contributes an additional term:

$$\dot{\varepsilon} = -\Gamma J^T \Gamma^{-1} \varepsilon + \Gamma \nabla (\nabla \cdot f).$$

The natural boundary condition still gives $\varepsilon(\tau) = 0$, but this equation is now forced. The zero residual need not solve it. This is why the unqualified claim “the most likely path always follows the drift” would be too strong.

The paper motivates neglecting this contribution by locally approximating the drift as linear over a suitable short interval, or by considering small noise. If $\Gamma = \epsilon D$ with fixed D , the quadratic term scales as $1/\epsilon$ while the divergence term is order one. This identifies a leading-order small-noise action; it does not establish a uniform correction estimate for every nonlinear problem.

2.15 14. Sanity checks and exercises

1. **No drift:** set $f = 0$, keep the endpoint free. What path minimizes S ?
2. **An imposed endpoint:** for $f = 0$ and $x(\tau) = b$, find the minimizing path and its action in one dimension.
3. **More noise:** replace scalar γ by 2γ while keeping candidate paths fixed. What changes about their quadratic costs and density ratios?
4. **A special endpoint:** in the linear bridge example, choose $b = x_0(1 - k\Delta t)^N$. What should happen to the discrete minimum?
5. **First variation:** why can we not discard the terminal boundary term in the free-endpoint problem?
6. **Convexity:** why does a quadratic cost in ε not prove convexity in x ?

2.15.1 Worked answers

1. The constant path $x(t) = x_0$ has zero residual and zero cost. Brownian samples nevertheless fluctuate.
2. The straight line $x(t) = x_0 + (b - x_0)t/\tau$ minimizes the integral of squared velocity. Its cost is $(b - x_0)^2/(4\gamma\tau)$. One proof is Cauchy–Schwarz: $(\int \dot{x} dt)^2 \leq \tau \int \dot{x}^2 dt$.

3. Every fixed-path quadratic cost halves. On the same grid, ratios become $\exp[-\Delta S/2]$ instead of $\exp[-\Delta S]$: likelihood differences become less pronounced. The common normalizer changes too, but cancels in ratios.
4. The imposed endpoint is now compatible with the Euler drift path, so the minimum returns to zero.
5. $\eta(\tau)$ is arbitrary, not zero. Its coefficient must vanish, giving the natural boundary condition. Forgetting it admits extraneous solutions of the interior Euler–Lagrange equation.
6. A nonlinear $f(x)$ makes the map $x \mapsto \varepsilon$ nonlinear. The composition need not be convex. The zero-cost proof remains valid because it attains a global lower bound.

As $\Gamma \rightarrow 0$ with a fixed admissible nonzero residual, the quadratic cost diverges. Under suitable regularity over finite times, the stochastic trajectories concentrate around the noiseless solution. This is a limiting statement about the stochastic dynamics; the absolute normalized action also contains grid-dependent constants.

2.16 15. Read Equation (4) once more

Line	Read it in words	What makes it work here?
$\mathbf{x}[\tau] = \arg \min \mathcal{A}$	Select a whole path with largest density	Negative logarithms reverse likelihood ordering
$\delta_x \mathcal{A} = 0$	No admissible infinitesimal deformation changes its action at first order	Interior Euler–Lagrange equation and free-terminal boundary condition
$\dot{\mathbf{x}} = f(\mathbf{x})$	The selected path follows the drift	Nonnegative quadratic cost can reach zero; divergence contribution omitted

The action scores a whole path; it is not the state surprisal at the endpoint. Minimizing it does not say that state surprisal decreases monotonically at every moment, and it does not require an organism consciously to predict anything.

This supplies a useful foundation for later variational arguments in the FEP. Later equations introduce different states and objectives; their assumptions still need checking individually.

2.17 Sources and using this notebook in a Quarto book

Primary source: Karl Friston, Lancelot Da Costa, Noor Sajid, Conor Heins, Kai Ueltzhöffer, Grigorios A. Pavliotis, and Thomas Parr (2023). *The free energy principle made simpler but*

not too simple. Physics Reports, 1024, 1–29. DOI: [10.1016/j.physrep.2023.07.001](https://doi.org/10.1016/j.physrep.2023.07.001). See Section 2, Equations (1)–(5), especially footnotes 3–4. This notebook develops the supplied teaching drafts while correcting the white-noise scaling and convexity claims. The finite-grid derivations, endpoint analysis, and numerical demonstrations are worked out here.

Quarto workflow: [Quarto’s JupyterLab guide](#) specifies a raw YAML cell for document options, Markdown cells for narrative, and `#|` comments for code-cell options. This notebook follows that structure. It has no external data or image dependencies.

Place `equation-4-least-action.ipynb` beside your other chapters and append it to your existing `book.chapters` list in `_quarto.yml`, preserving the other entries:

```
book:
  chapters:
    - index.qmd
    # Keep your other chapter entries here.
    - equation-4-least-action.ipynb
```

The `index.qmd` line is illustrative: keep your actual existing index and configuration. See [Quarto book structure](#).

Install the runtime packages if needed:

```
python -m pip install jupyterlab numpy scipy matplotlib
```

Run all cells in JupyterLab and save. Quarto normally uses the saved notebook outputs. From a terminal in your book directory:

```
quarto preview
```

To render and recompute this notebook explicitly:

```
quarto render equation-4-least-action.ipynb --execute
```

The notebook uses seeded simulations for reproducibility. Monte Carlo intervals in the nonlinear example quantify sampling uncertainty only; they do not bound Euler time-discretization error.

3 From sparse dynamics to Markov blankets

Equation (11), with conceptual connections to Andy Clark

3.1 Why this small equation matters

Equation (11) connects two different descriptions of a system: **how states influence each other’s motion**, and **what knowing one state tells us about another**. Under the paper’s particular sparse-coupling assumptions, missing dynamical influences lead to conditional independencies. Those independencies support a partition into internal, external, sensory, and active states.

We will follow the original explanation step by step: read the equation; understand its Jacobian and Hessian; unpack conditioning; read each line; prove the probability equivalence; and return to the question of what makes something a “thing.”

Along the way, Andy Clark’s *How to Knit Your Own Markov Blanket: Resisting the Second Law with Metamorphic Minds* (2017) helps us interpret the boundary. Clark argues that Markov blankets are compatible with embodied, enactive, and extended cognition because relevant boundaries can be multiple and change over time. That is his philosophical argument, not a theorem supplied by Equation (11).

Prerequisites: partial derivatives, joint and conditional densities, and elementary matrix multiplication. The numerical example uses NumPy, SciPy, and Matplotlib. All its outputs are embedded in this notebook; it needs no external data files.

Scope: p below is the paper’s smooth positive steady-state density. Zeros such as $H_{\mu\eta} = 0$ mean a block vanishes throughout the relevant domain, not merely at one point. For the converse from zero mixed derivatives to independence, assume a connected product support in the variables being separated, with well-defined normalizable conditional densities. These qualifications matter for constrained or disconnected supports.

3.2 1. Read the complete equation

$$\begin{aligned}
\mathbf{J}_{\mu\eta} = 0 &\Rightarrow \mathbf{H}_{\mu\eta} = 0 \Leftrightarrow (\mu \perp \eta) \mid b \\
\mathbf{J}_{a\eta} = 0 &\Rightarrow \mathbf{H}_{a\eta} = 0 \Leftrightarrow (a \perp \eta) \mid s, \mu \\
\mathbf{J}_{s\mu} = 0 &\Rightarrow \mathbf{H}_{s\mu} = 0 \Leftrightarrow (s \perp \mu) \mid a, \eta.
\end{aligned} \tag{11}$$

This is Equation (11) in Friston et al. (2023), Section 4. The notation is:

Symbol	Role in the partition
η	External states
s	Sensory states
a	Active states
μ	Internal states
$b = (s, a)$	Blanket states

Each symbol can denote several variables. Hence the entries of $\mathbf{J}$ and $\mathbf{H}$ can be blocks rather than scalars.

The central chain is

$$\text{missing direct influence} \xRightarrow{\text{special sparse coupling}} \text{zero mixed surprisal curvature} \Leftrightarrow \text{conditional independence.}$$

The qualification over the first arrow is part of the result. It is not valid for arbitrary stochastic dynamics.

3.3 2. Start with the Jacobian: which way does influence go?

For a drift field $f(x)$, define

$$J_{uv} = \frac{\partial f_u}{\partial x_v}.$$

Hold all other state coordinates fixed and slightly change x_v . Does the instantaneous drift of x_u change? That is what this derivative asks.

For example,

$$J_{\mu\eta} = \frac{\partial f_\mu}{\partial \eta} = 0$$

means that external states do not directly enter the internal drift, over the domain where this equality holds.

The second index influences the first. $J_{\mu\eta}$ concerns $\eta \rightarrow \mu$, not $\mu \rightarrow \eta$.

Take a toy drift $f_\mu = -\mu + 2s$. It depends directly on μ and s , but not η . Nevertheless, if η influences s , changes outside can propagate to the internal variable over time. No direct edge does not mean no indirect effect.

Clark connection — statistical versus causal language. In Section 1, Clark describes a blanket as an informational screening relation. In footnote 11 he stresses its statistical rather than intrinsically causal definition. Here we begin with a *dynamical* object, J , and need additional assumptions to reach the *statistical* object. Keeping the two apart prevents the misleading inference that a sparse-looking interaction diagram automatically proves a blanket.

3.4 3. The Hessian: does one slope depend on another state?

Let steady-state surprisal be

$$\mathfrak{J}(x) = -\ln p(x).$$

Its Hessian is

$$H_{uv} = \frac{\partial^2 \mathfrak{J}}{\partial x_u \partial x_v}.$$

First differentiate with respect to μ : how does surprisal change when the internal coordinate changes? Then differentiate that slope with respect to η : does changing the external coordinate alter the slope?

$$H_{\mu\eta} = \frac{\partial}{\partial \eta} \left(\frac{\partial \mathfrak{J}}{\partial \mu} \right).$$

If this block is identically zero, the internal slope has no dependence on the external coordinate while the remaining variables are held fixed.

For an elementary example,

$$\mathfrak{I}(\mu, \eta \mid b) = \frac{1}{2}(\mu - m(b))^2 + \frac{1}{2}(\eta - e(b))^2 + C(b)$$

has zero mixed derivative. Add a term $c\mu\eta$, and the mixed derivative becomes c . The additional term ties the two coordinates together even with b fixed.

Unlike the Jacobian, a smooth scalar function's Hessian is symmetric: $H_{uv} = H_{vu}^\top$. A directed missing dynamical influence can therefore lead, under the stated assumptions, to a symmetric conditional-independence relation.

3.5 4. What does conditioning on a blanket mean?

The symbol $\perp$ denotes statistical independence. The vertical bar names what is already known:

$$\mu \perp \eta \mid b, \quad b = (s, a).$$

Equivalently,

$$p(\mu, \eta \mid s, a) = p(\mu \mid s, a)p(\eta \mid s, a),$$

or

$$\boxed{p(\mu \mid \eta, s, a) = p(\mu \mid s, a)}.$$

Once the sensory and active states are specified, additionally learning the external states does not change the conditional distribution of the internal states.

This is not a claim that μ is known exactly: $p(\mu \mid b)$ can remain broad. It is not a claim that the future of μ is perfectly predictable from a single snapshot of b either.

Clark connection — sufficient information, not perfect prediction. Section 1 characterizes a blanket in terms of which other variables are statistically informative. Footnote 2 explicitly qualifies this: intrinsic dynamics mean that knowing the blanket does not give perfect prediction of the contained system's evolution. Our equation is a statement about a conditional state distribution; path-level statements require further dynamical arguments.

Clark also distinguishes a **Markov blanket**, a screening set, from a **Markov boundary**, a minimal screening set. Equation (11)'s first line establishes screening by (s, a) ; minimality is a separate issue.

3.6 5. The translator analogy

Imagine Alice’s statement A , a translator’s output T , and Bob’s understanding B . Suppose Bob’s understanding depends only on the translation and independent personal noise:

$$p(A, T, B) = p(A)p(T | A)p(B | T).$$

Then $B \perp A | T$. Without the translation, Alice’s statement can tell us a great deal about Bob’s understanding. Once the translation is known, her original words add nothing to the distribution specified by this model.

The assumption is important: if Bob can also see Alice’s gestures, or shares an unmodeled context with her, the screening may fail. A story about an intermediary is not enough; the factorization must hold.

The blanket plays this screening role for internal and external states. Unlike this simple one-way analogy, an FEP partition can participate in reciprocal sensing and acting over time.

3.7 6. First line: internal and external states

Read the first line as three steps:

$$J_{\mu\eta} = 0 \Rightarrow H_{\mu\eta} = 0 \Leftrightarrow \mu \perp \eta | (s, a).$$

1. **Dynamics:** external states do not directly change the internal drift.
2. **Sparse-coupling argument:** the paper restricts the terms relating J to H , excluding cancellation of nonzero contributions. In that setting, the mixed Hessian block must vanish.
3. **Probability:** the zero block expresses factorization of the internal–external conditional density given the blanket.

This is the familiar internal/blanket/external separation. It does not establish $\mu \perp \eta$ without conditioning.

Clark connection — a boundary that supports exchange. Section 2 places blanket talk within the problem of organisms maintaining characteristic conditions through environmental exchanges. Such a boundary need not be an impermeable wall. A cell’s receptors and effectors can participate in the processes that maintain its organization. Assigning concrete biological variables to (η, s, a, μ) , however, is a modeling decision whose independence claims must be justified.

Equation (11) alone supplies no law of survival, thermodynamic entropy production, or conscious experience. Clark’s persistence story motivates why such partitions matter to the FEP; additional theory is needed to connect the partition to those broader claims.

3.8 7. Why the organism is still connected to the world

Consider the schematic sequence

$$\text{external temperature} \rightarrow \text{receptor state} \rightarrow \text{internal response}.$$

The internal response can correlate strongly with temperature, even though the model routes the influence through receptor states. Conditional independence asks what remains informative **after the appropriate mediating variables are specified**.

In a general dynamical system, history and shared causes can complicate this story. We cannot infer snapshot conditional independence merely from the existence of a mediating route. The paper’s steady-state and sparse-coupling construction supplies the extra mathematical structure.

Clark connection — embodied and extended cognition. In Sections 3–4, Clark argues that bodily and environmental resources can participate in cognitive activity despite statistical boundaries. The screening equation does not decide whether a gesture, tool, or notebook is constitutive of a cognitive process rather than merely causally helpful. Clark treats that as a further explanatory and philosophical question. “There is a blanket” and “the mind ends exactly here” are different claims.

3.9 8. Second line: external states and active states

$$J_{a\eta} = \frac{\partial f_a}{\partial \eta} = 0 \Rightarrow H_{a\eta} = 0 \Leftrightarrow a \perp \eta \mid (s, \mu).$$

The environment does not enter the active drift directly in this partition. It can still change activity through sensory and internal states. A schematic permitted route is

$$\eta \rightarrow s \rightarrow \mu \rightarrow a.$$

The conditional statement is

$$p(a \mid \eta, s, \mu) = p(a \mid s, \mu).$$

This is a definition of roles within the chosen architecture, not a universal claim about everything ordinarily called an action. For example, a limb being pushed by an external force may need a different variable assignment than the abstract “active states” used here.

3.10 9. Why condition on s, μ in that line?

The Hessian criterion tests two variable groups while holding the remaining groups fixed. For this four-part state description:

Pair tested	Remaining groups to condition on
μ, η	s, a
a, η	s, μ
s, μ	a, η

The conditioning set is part of the claim. Changing or removing a conditioning variable can change whether independence holds. In particular, conditioning is not the same operation as marginalizing: knowing a variable and averaging over an unknown variable generally produce different distributions.

The numerical example below will show this explicitly: a covariance entry can be nonzero even when the corresponding precision/Hessian entry is zero.

3.11 10. Third line: internal states and sensory states

$$J_{s\mu} = \frac{\partial f_s}{\partial \mu} = 0 \Rightarrow H_{s\mu} = 0 \Leftrightarrow s \perp \mu \mid (a, \eta).$$

Internal states do not directly enter the sensory drift. They can nevertheless affect future sensory states through action. Turning toward a sound changes what the sensors encounter; no direct $\mu \rightarrow s$ term is required in the chosen model.

For example, an allowed route is $\mu \rightarrow a \rightarrow s$; another can run through external states. These are schematic dependencies, not a claim that biological sensory systems never receive feedback. A proposed mapping to retina, cortex, muscles, or receptors must fit the architecture rather than inherit it simply from anatomical labels.

Clark connection — changing the sensory interface. In Section 5, Clark uses examples such as spectacles and augmented sensory devices to question the assumption of one permanently fixed boundary. Mathematically, this invites us to revisit which variables belong to s , a , and the other groups. It does not permit assigning a new blanket arbitrarily: the relevant conditional-independence relations must hold for the revised model.

3.12 11. Put the three lines together

Equation (11) excludes three particular direct influences: $\eta \rightarrow \mu$, $\eta \rightarrow a$, and $\mu \rightarrow s$. A familiar schematic perception–action cycle allowed within the subsequent architecture is

$$\eta \rightarrow s \rightarrow \mu \rightarrow a \rightarrow \eta.$$

This is only a subset of possible dynamical dependencies, not a conditional-independence graph. Equations (12)–(13) supply a fuller architecture. The three displayed restrictions in Equation (11) should not be treated as the entire specification of every allowable edge.

For the Gaussian example we will use an **undirected statistical graph**: η linked to s , s linked to a , and a linked to μ . An edge in that graph denotes nonzero precision, not an arrow of dynamical influence. It gives a simple distribution satisfying all three conditional-independence statements.

3.13 12. Why the two arrows are different

The first arrow is conditional on the paper’s dynamical assumptions. Equation (8) gives

$$f = \Omega \nabla \mathfrak{J} - \Lambda, \quad J = \Omega H + \nabla \Omega \cdot \nabla \mathfrak{J} - \nabla \Lambda,$$

where $\Omega = Q - \Gamma$, $Q = -Q^\top$, and $\Lambda_i = \sum_j \partial_j \Omega_{ij}$ in the paper’s setting. In components,

$$J_{uv} = \sum_i \Omega_{ui} H_{iv} + \sum_i (\partial_v \Omega_{ui}) (\partial_i \mathfrak{J}) - \sum_i \partial_i \partial_v \Omega_{ui}.$$

A zero sum does not normally mean every term is zero. Equation (9) defines a restricted sparse-coupling construction in which the relevant contributions vanish individually. Among its requirements are $\Gamma_u H_{uv} = 0$ and vanishing derivatives of the relevant Ω entries. With invertible positive diffusion in that block, $\Gamma_u H_{uv} = 0$ forces $H_{uv} = 0$.

Thus the notation $J_{uv} = 0 \Rightarrow H_{uv} = 0$ in Equation (11) must be read **within that construction**, not as a deduction from an observed zero Jacobian alone. The solenoidal restrictions and derivative conditions cannot be dropped silently.

The second arrow, $H_{uv} = 0 \Leftrightarrow x_u \perp x_v \mid \text{others}$, follows from probability factorization under the smoothness, positivity, and support conditions stated at the beginning. We now prove both directions.

3.14 13. The probability argument behind the zero Hessian

3.14.1 Independence implies zero mixed curvature

Suppose $\mu \perp \eta \mid b$. Then

$$p(\mu, \eta, b) = p(\mu \mid b)p(\eta \mid b)p(b).$$

Taking negative logarithms turns the product into an additive separation:

$$\mathfrak{J}(\mu, \eta, b) = \mathfrak{J}(\mu \mid b) + \mathfrak{J}(\eta \mid b) + \mathfrak{J}(b).$$

No term jointly involves μ and η . Differentiating once with respect to each gives $H_{\mu\eta} = 0$.

3.14.2 Zero mixed curvature implies independence

Now assume $\partial_\eta \partial_\mu \mathfrak{J} = 0$ **everywhere** on a connected product domain at fixed b . The μ -slope is independent of η . Integrating that fact gives

$$\mathfrak{J}(\mu, \eta, b) = U(\mu, b) + V(\eta, b),$$

where an arbitrary b -only term can be included in either function. Exponentiating,

$$p(\mu, \eta, b) = e^{-U(\mu, b)} e^{-V(\eta, b)}.$$

At fixed b , the normalization separates into $Z_\mu(b)Z_\eta(b)$ because the support is a product. Therefore

$$p(\mu, \eta \mid b) = \frac{e^{-U(\mu, b)}}{Z_\mu(b)} \frac{e^{-V(\eta, b)}}{Z_\eta(b)} = p(\mu \mid b)p(\eta \mid b).$$

The same argument applies to the other two pairs. A derivative that happens to vanish at a single state does not establish this factorization.

3.14.3 See it numerically: correlated outside and inside, independent given the blanket

For a centered multivariate Gaussian,

$$p(x) \propto \exp(-\frac{1}{2}x^T K x), \quad H = K = \Sigma^{-1}.$$

Here K is the precision and Σ the covariance. Choose a positive-definite tridiagonal precision in the order (η, s, a, μ) . It has exactly the three off-diagonal zeros in Equation (11).

For the unknown pair $U = (\eta, \mu)$ and observed blanket $B = (s, a)$,

$$\text{Cov}(U \mid B) = \Sigma_{UU} - \Sigma_{UB} \Sigma_{BB}^{-1} \Sigma_{BU}.$$

That conditional covariance will be diagonal even though the marginal covariance Σ_{UU} is not. Gaussianity makes zero conditional covariance equivalent to conditional independence here; zero correlation alone would not suffice for arbitrary non-Gaussian distributions.

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.linalg import solve_continuous_lyapunov

plt.rcParams.update({"figure.dpi": 120, "font.size": 11,
                    "axes.spines.top": False, "axes.spines.right": False})
labels = [r"$\eta$", r"$s$", r"$a$", r"$\mu$"]
K = 2*np.eye(4) - 0.9*np.eye(4, k=1) - 0.9*np.eye(4, k=-1)
assert np.linalg.eigvalsh(K).min() > 0
Sigma = np.linalg.inv(K)
U, B = [0, 3], [1, 2]
Suu = Sigma[np.ix_(U, U)]
Sub = Sigma[np.ix_(U, B)]
Sbb = Sigma[np.ix_(B, B)]
conditional_cov = Suu - Sub @ np.linalg.solve(Sbb, Sub.T)
assert abs(conditional_cov[0, 1]) < 1e-12
assert all(K[i,j] == 0 for i,j in [(3,0),(2,0),(1,3)])
# A compatible equilibrium diffusion: Q=0, Gamma=0.5 I, f=Jx.
Gamma = 0.5*np.eye(4)
J = -Gamma @ K
assert np.allclose(J @ Sigma + Sigma @ J.T + 2*Gamma, 0)
assert np.max(np.linalg.eigvals(J).real) < 0
marginal_corr = Suu[0,1]/np.sqrt(Suu[0,0]*Suu[1,1])
print("Marginal correlation of external and internal states:", round(marginal_corr,4))
```

```

print("Conditional covariance given sensory and active states:")
print(np.round(conditional_cov, 6))
print("All three precision zeros and the stationary covariance equation verified.")

```

```

Marginal correlation of external and internal states: 0.1532
Conditional covariance given sensory and active states:
[[0.5 0. ]
 [0.  0.5]]
All three precision zeros and the stationary covariance equation verified.

```

```

fig, axes = plt.subplots(1, 3, figsize=(12, 3.8), constrained_layout=True)
for ax, matrix, title in zip(axes[:2], [K, Sigma], ["Precision = surprisal Hessian", "Covariance = surprisal Hessian inverse"]):
    im = ax.imshow(matrix, cmap="RdBu_r", vmin=-2, vmax=2)
    ax.set_xticks(range(4), labels); ax.set_yticks(range(4), labels)
    ax.set_title(title, fontsize=11)
    for i in range(4):
        for j in range(4):
            ax.text(j, i, f"{matrix[i,j]:.2f}", ha="center", va="center",
                    color="white" if abs(matrix[i,j]) > 1.3 else "black", fontsize=9)
# Draw analytic covariance ellipses, not regression-based approximations.
angle = np.linspace(0, 2*np.pi, 250)
circle = np.vstack([np.cos(angle), np.sin(angle)])
for cov, color, label in [(Suu, "#a23b72", "Marginal"),
                          (conditional_cov, "#087e8b", "Conditional, b = 0")]:
    ellipse = np.linalg.cholesky(cov) @ circle
    axes[2].plot(*ellipse, color=color, label=label)
axes[2].set(xlabel="External state", ylabel="Internal state", title="Equal Mahalanobis-radius ellipses")
axes[2].set_aspect("equal"); axes[2].legend(fontsize=8)
plt.show()

```

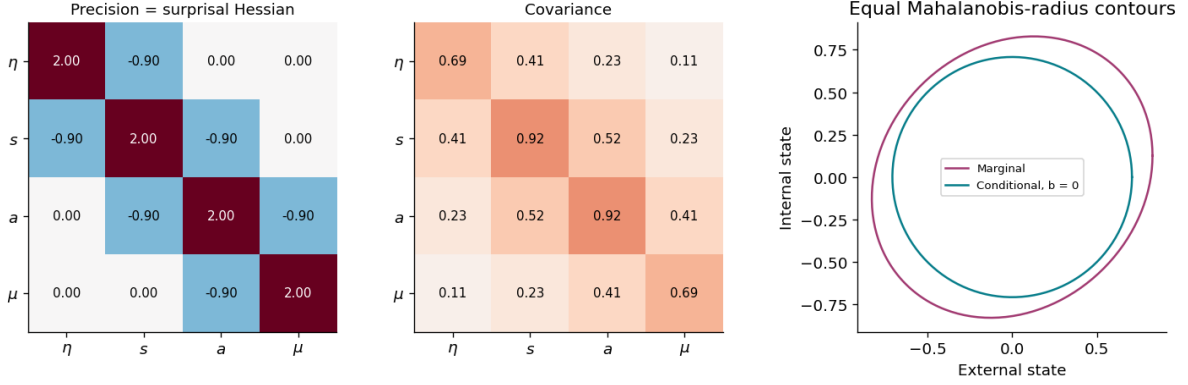


Figure 3.1: The precision has the required zeros, although the covariance connects distant variables. Fixing the blanket removes the external–internal conditional correlation.

The compatible diffusion in the code deliberately takes $Q = 0$ and constant diagonal noise, so $J = -\Gamma H$. The implication from zero J to zero H is transparent in this restricted example. It is a valid special case, not a simulation of a complete biological perception–action system.

3.14.4 A counterexample if we drop the special sparse-coupling assumptions

Consider two variables with drift $f(x) = Ax$,

$$A = \begin{bmatrix} -1 & 0 \\ 1 & -1 \end{bmatrix}, \quad \Gamma = I.$$

There is no direct influence of variable 2 on the drift of variable 1: $A_{12} = 0$. Nevertheless, variable 1 drives variable 2, producing a correlated stationary density. The stationary covariance solves

$$A\Sigma + \Sigma A^\top + 2I = 0.$$

The next cell checks that its precision has $H_{12} \neq 0$, and exhibits the cancellation excluded by the paper’s special definition.

```
A = np.array([[ -1.,  0.], [ 1., -1.]])
Sigma_c = solve_continuous_lyapunov(A, -2*np.eye(2))
H_c = np.linalg.inv(Sigma_c)
Q_c = A @ Sigma_c + np.eye(2)
Omega_c = Q_c - np.eye(2)
assert np.allclose(Q_c, -Q_c.T)
```

```

assert np.allclose(Omega_c @ H_c, A)
assert A[0,1] == 0 and abs(H_c[0,1]) > 0.1
terms = Omega_c[0,:]*H_c[:,1]
assert abs(terms.sum()) < 1e-12 and np.all(np.abs(terms)>0.1)
print("Stationary covariance:\n", Sigma_c)
print("Steady-state Hessian (precision):\n", H_c)
print("Contributions to J_12 = sum_i Omega_1i H_i2:", terms)
print("They cancel, giving J_12 = 0 despite H_12 != 0.")

```

Stationary covariance:

```

[[1.  0.5]
 [0.5 1.5]]

```

Steady-state Hessian (precision):

```

[[ 1.2 -0.4]
 [-0.4  0.8]]

```

Contributions to J_12 = sum_i Omega_1i H_i2: [0.4 -0.4]

They cancel, giving J_12 = 0 despite H_12 != 0.

This does not contradict Equation (11) as qualified by Equations (9)–(10): the example violates the no-cancellation restriction. It shows precisely why the qualification is needed.

3.15 14. The conceptual chain from Equation (6) to Equation (11)

Step	Mathematical statement	Question answered
Equation (6)	$f = \Omega \nabla \mathfrak{J} - \Lambda$	How is drift related to a steady-state probability landscape?
Equation (8)	$J = \Omega H + \nabla \Omega \cdot \nabla \mathfrak{J} - \nabla \Lambda$	How are dynamical sensitivities related to mixed curvature?
Equations (9)–(10)	Relevant terms vanish individually	Which restrictions make missing influences imply zero curvature?
Equation (7)	$H_{uv} = 0 \Leftrightarrow x_u \perp x_v \mid \text{others}$	How does curvature encode screening?
Equation (11)	Apply this to the three pairs	Which independencies support the particular partition?

Clark connection — FEP is broader than predictive processing. In Section 2 and footnotes 6–9, Clark distinguishes describing an adaptive system as a self-evidencing model from claiming that it implements online top-down prediction-error minimization. That distinction belongs here: none of the derivatives in Equation (11) establishes a predictive-coding algorithm. Later FEP constructions may support an inference interpretation under further assumptions, but this equation alone provides a partition and statistical constraints.

3.16 15. Why this matters for the idea of a “thing”

The mathematical strategy begins with variables and relations rather than presupposing a membrane, skull, or organism. A blanket gives a statistical way to distinguish an internal set from a remaining external set. That is useful for studying how an organized system relates to its surroundings.

There is a subtlety specific to this paper. Section 4 constructs **active states as a Markov boundary of internal states**, then **sensory states as a Markov boundary of autonomous states** $\alpha = (a, \mu)$. Their union $b = (s, a)$ is called the blanket. Consequently, b should not automatically be called the *minimal* boundary of μ . In our Gaussian chain, a alone screens μ from (η, s) , while (s, a) also screens it from η . This is a concrete example of the blanket/boundary distinction Clark makes in Section 1.

3.16.1 Multiple blankets, without arbitrary boundaries

Clark’s Section 1 discusses nested organizations; Sections 5–6 develop the argument that biological and cognitive boundaries can be multiple and malleable. Different internal sets, scales, or explanatory questions can identify different useful screening relations. This should not be read as proving multiple minimal boundaries for one fixed target in one fixed positive distribution, nor that adding arbitrary variables always preserves screening.

Our tridiagonal Gaussian gives a small exact illustration: a is a boundary for μ , while s screens the pair (a, μ) from η . The nested statistical partition is demonstrable. Identifying the corresponding variables in a real organism is a separate empirical task.

3.16.2 Metamorphosis and a process view of identity

In Section 5 Clark uses the caterpillar–pupa–butterfly lifecycle to argue that persistence need not require one unchanging morphology. Section 6 develops a process-based account: learning and development can reorganize the resources through which an agent senses, acts, and maintains itself.

This is a productive interpretation of the mathematics, but Equation (11) concerns a specified partition and the paper’s steady-state description. It does **not** itself prove that the same density and exact independence relations remain valid throughout metamorphosis. Modeling such change requires an expanded state description, a justified separation of timescales, or a time-dependent model with its assumptions checked anew.

Clark’s central philosophical proposal is therefore compatible with, but not entailed by, our derivation: boundaries can matter without one particular anatomical boundary settling every question about cognition.

3.16.3 A question to carry into the rest of the book

When someone proposes a Markov blanket, ask: **Which variables, which internal target, which conditioning set, which probability distribution, and which timescale?** Those choices make the claim testable and keep the mathematical screening relation connected to the explanatory purpose.

3.17 Check your understanding

1. What influence does $J_{s\mu}$ measure?
2. Does $H_{\mu\eta}(x_*) = 0$ at one point prove independence?
3. Can $\Sigma_{\mu\eta} \neq 0$ coexist with $H_{\mu\eta} = 0$ for a Gaussian?
4. What extra premise prevents the two-variable counterexample from invalidating the paper’s qualified implication?
5. Does knowing a blanket determine an internal state exactly?
6. Does Clark’s argument prove that a particular smartphone lies inside an exact Markov blanket?

3.17.1 Answers

1. The sensitivity of sensory drift to internal states: the second index influences the first.
2. No. We need the mixed block to vanish over the relevant domain and suitable density/support conditions.
3. Yes. Marginal correlation and conditional dependence are different. Precision zeros encode Gaussian conditional independence given the other variables.
4. The paper imposes termwise vanishing, excluding cancellation of nonzero contributions and constraining the solenoidal terms.
5. No. It specifies a conditional distribution; uncertainty and intrinsic dynamics remain. See Clark, footnote 2.

6. No. Clark argues that cognition need not have a uniquely fixed biological boundary. A proposed extended system still requires a defensible model and evidence for its statistical and functional organization.

The sentence to remember is: **Under the paper’s special sparse-coupling assumptions, absent direct influences imply zero mixed curvature of steady-state surprisal; those zeros express the conditional independencies defining the partition.**

3.18 Sources and Quarto setup

Mathematical source: Friston, K., Da Costa, L., Sajid, N., Heins, C., Ueltzhöffer, K., Pavliotis, G. A., and Parr, T. (2023). *The free energy principle made simpler but not too simple*. Physics Reports, 1024, 1–29. DOI. Section 4, Equations (7)–(13), especially the definition in (9), footnote 10, and the partition construction preceding (11). Checked against the uploaded PDF.

Conceptual source: Clark, A. (2017). *How to Knit Your Own Markov Blanket: Resisting the Second Law with Metamorphic Minds*. Uploaded EPUB, authored content OEBPS/03_Clark.xhtml. Citations in this notebook use its numbered sections and footnotes because EPUB page numbers vary by reader. Relevant locations: Section 1 and footnote 2 (screening, prediction, blanket/boundary); Section 2 and footnotes 6–9 (FEP versus predictive processing); Sections 3–4 (embodiment and extension); Sections 5–6 (metamorphosis, malleability, process); Section 8 (conclusion). Clark’s views are paraphrased and attributed; no chapter text or illustrations are reproduced.

Teaching source: the supplied Pasted markdown(5).md. This notebook retains its explanatory sequence while adding the converse proof, numerical illustrations, source connections, and qualifications on sparse coupling and support. Clark’s discussion predates the 2023 paper; the connections drawn here are pedagogical comparisons, not claims that Clark comments on Equation (11).

Save this notebook as `chapters/equation-11-markov-blankets.ipynb` in your book. Add this entry to your existing book.chapters list in `_quarto.yml`:

```
- chapters/equation-11-markov-blankets.ipynb
```

The raw first cell contains Quarto metadata. All computations are self-contained and outputs are embedded. To recompute, install `numpy` `scipy` `matplotlib` in your Jupyter environment and run the code cells in order, then save. From the book root, use `quarto preview`, or explicitly recompute with:


```
quarto render chapters/equation-11-markov-blankets.ipynb --execute
```

See [Quarto's JupyterLab guide](#). The source EPUB and PDF are cited reading materials and are not runtime dependencies.

4 Summary

In summary, this book has no content whatsoever.

References

Knuth, Donald E. 1984. “Literate Programming.” *Comput. J.* (USA) 27 (2): 97–111.
<https://doi.org/10.1093/comjnl/27.2.97>.